Author Response to Reviewers

AFE(2026)5804, resubmitted as a new manuscript

Dear Dr Ranger and Dr MacQuarrie,

Thank you for the decision of 22 September 2026 and for the invitation to resubmit. Each comment is quoted below as it appeared in the decision email, followed by what was done and where in the revised manuscript it can be found. Section numbers refer to the revised manuscript. Two defects in the submitted data were repaired during the revision, the wind terms were rebuilt, and several findings changed as a result. The repairs, and which findings held and which changed, are set out after the responses so that nothing in the new manuscript arrives unannounced.

Editor-in-Chief

Comment 1. “I agree with the Associate Editor that the manuscript is within the scope of AFE, but the Introduction and Discussion need to be thoroughly revised to include existing literature.”

Response. The literature was gathered in two searches under written protocols, on 23 and 24 September 2026, which screened 1,257 and 4,430 unique records, and the reference list of the main text now holds 61 works against 30 in the submitted manuscript, with the sources of the wind and heat models moved to the Supporting Information. The Introduction (section 1) was rewritten on that literature, one study to a sentence. It opens with the beetle as a native disturbance agent of pine forests and the history of its outbreaks (Macias Fauria & Johnson, 2009; Raffa et al., 2008; Safranyik & Carroll, 2006), defines disturbance refugia and the three mechanisms proposed for them (Cartwright, 2018; Krawchuk et al., 2020), and then sets out the biology each mechanism rests on, how a mass attack overcomes the defences of a tree and the threshold at which it does so (Boone et al., 2011; Raffa & Berryman, 1983), why large trees are the better hosts (Carroll & Safranyik, 2004; Safranyik et al., 2010), how beetles fly within a stand and how attack spreads from the year before (Aukema et al., 2008; Kunegel-Lion & Lewis, 2020; Robertson et al., 2007; Safranyik et al., 1989; Safranyik et al., 1992), how cold and drought act (Chapman et al., 2012; Logan & Powell, 2001; Preisler et al., 2012), how terrain entered earlier models (Nelson et al., 2007; Walter & Platt, 2013; Wulder et al., 2006), what thinning trials found and why (Amman et al., 1988; Bartos & Amman, 1989; Fettig et al., 2007; Mitchell et al., 1983;

Shore et al., 2006), and the two ways wind can act (Ainslie & Jackson, 2010; de la Giroday et al., 2011; Hynum & Berryman, 1980; Jackson et al., 2008). The Discussion (section 4) sets each finding against that literature, as described under comments 4 and 5.

Comment 2. “In addition, the novelty of the study needs to be explicitly stated since numerous modelling papers are submitted to AFE and we have limited room.”

Response. The novelty is stated in three places, as what the study did. The first statement of the abstract names the gap, how little is known about how the three mechanisms act together once attack nearby and earlier is taken into account. The last paragraph of the Introduction states that few studies had tested these mechanisms together while accounting for attack nearby, and sets the three questions. The opening paragraph of the Discussion states that the novelty of the study lay in testing the three mechanisms of Krawchuk et al. (2020) together on one landscape at 30 m with attack in and around each cell the year before in every model, and in testing the two ways wind could act, breaking up the pheromone plume and depositing beetles on sheltered ground, with a terrain wind model over the afternoon flight hours of each sixteen-day period.

Associate Editor

Comment 3. “I am, however, concerned that the arguments in the introduction and discussion are not well supported or linked to the (admittedly vast) literature on MPB.”

Response. Every argument in the Introduction now rests on named studies of this beetle, as set out under comment 1, and each of the three questions is derived from them. The first question follows the mechanisms proposed by Krawchuk et al. (2020) and the refugia Cartwright (2018) mapped in southern Oregon, with the thinning trials behind stand density and the 25 cm diameter at which Carroll & Safranyik (2004) placed the change from beetle sink to beetle source behind host size. The second follows Bartos & Amman (1989), who measured more light, warmth and wind in a thinned stand than in the unthinned stand beside it. The third follows the finding of Hynum & Berryman (1980) that pioneer beetles could not tell hosts from non-hosts while landing, and the settlement of beetles where wind slows (de la Giroday et al., 2011). The terms for attack nearby and in the previous year, entered in every model, follow Aukema et al. (2008), Robertson et al. (2007) and Kunegel-Lion & Lewis (2020). The Discussion rests on the same literature, section by section.

Comment 4. “My initial impression is that the Discussion reiterates what has already been stated in the results.”

Response. The Discussion (section 4) was rewritten in four sections that mirror the three questions of the Results, and coefficients are repeated only where an argument needs them. Its opening paragraph reads the dominance of attack nearby and the year before against Aukema et al. (2008) on the Chilcotin Plateau, the 140 m over which previous attack raised attack in Colorado (Walter & Platt, 2013) and the late phase of the outbreak (Meddens & Hicke, 2014). Section 4.1 reads the density result against the stand susceptibility rating used in British Columbia (Safranyik et al., 2010; Shore et al., 2000), the thinning trials and the history of the province’s pine (Taylor & Carroll, 2003), reads the shade result against the observation that the heaviest attacks on a bole are found on its northern side (Safranyik & Carroll, 2006) and that attacks fall at midday with the highest temperatures (Schmid et al., 1991), and reads host size against the dependence of brood production on phloem thickness (Amman, 1972), the 25 cm boundary between beetle sinks and sources (Carroll & Safranyik, 2004) and the depletion of large pine by epidemics (Shore et al., 2006). Section 4.2 reads the wind result against the stand measurements of Bartos & Amman (1989) and Thistle et al. (2004), wind above the canopy carrying beetles into new stands (Chen & Jackson, 2017; Jackson et al., 2008) and pheromones from adjacent stands overriding the benefit of spacing (Schmid & Mata, 2005). Section 4.3 reads the shelter result against the windward attack de la Giroday et al. (2011) found in the Peace River region, the crosswind and downwind flight that Safranyik et al. (1989) recorded, the flight limit of Carroll & Safranyik (2004), the north-facing onset in the Cypress Hills (Kunegel-Lion & Lewis, 2020), deposition within the stand (Byers, 2000) and the scale of inference that Aukema et al. (2008) raised. Section 4.4 sets out the limits of the study and what the findings mean for management.

Comment 5. “Their revised discussion should link the author’s findings to the biology, ecology or management of the MPB system.”

Response. Each section of the Discussion reads a result through the beetle rather than through the model. The biology appears in sections 4.1 and 4.3, through the brood that thicker phloem feeds, the side of the bole that is attacked, the fall in attack at midday heat, flight crosswind and downwind of an attacked tree and the wind speed above which the beetle cannot fly. The ecology appears in sections 4.1 and 4.3, through the age structure of the province’s pine, the pine that grew on the higher ground of this range, the phase of the outbreak and the difference between a refugium defined at 30 m and one defined at 1 km. Management appears in sections 4.2 and 4.4. Section 4.2 states that a stand opened by thinning was protected by the wind only during flight

and least where attacked trees stood close, and section 4.4 states that the refugia this landscape showed were stands with little host, the structure that thinning and a younger age structure produce, which terrain maps alone could not locate, and sets that against the rating under which partial cutting lowers the susceptibility of a stand (Mata et al., 2003; Shore et al., 2000), the advice to manage on a landscape basis because small cut parcels surrounded by unmanaged stands may not be protected (Schmid & Mata, 2005), and the conclusion of Six et al. (2014) that the field evidence for thinning during an outbreak was weaker than policy had assumed.

Analysis repairs

Two defects in the submitted data were repaired, the wind terms were rebuilt, and the spatial statistics of Aukema et al. (2008) were added. The design was kept, with annual maps for the first and third questions and sixteen-day maps for the second.

1. The attack maps were rebuilt with a new classifier, trained on the 112 Landsat pixels of the field plots, four to a plot, against undisturbed forest digitised for the companion study, and tested over 100 random splits that each held out a quarter of the plots and of the 100 m blocks of undisturbed pixels. The radial support vector machine reached kappa 0.859 ± 0.081, and its cut was applied to the annual maps and to the 47 sixteen-day maps alike (section 2.2, Table S6). The table of attack by diameter class was rebuilt on the repaired maps (Table S8). The annual maps in the submitted manuscript came from an earlier classifier trained on 38 plots, with a leave-one-out kappa of 0.475, while its Methods described the classification of Murphy et al. (2026). The summer of 2012, which the submitted manuscript left out, was mapped from Landsat 7 alone and entered the annual models (section 2.2).
2. The terrain-adjusted station wind was recomputed on an elevation model extending beyond the study area. In the submitted analysis its curvature term was computed on an elevation model cut to the study area, so the field covered 18,388 of the 61,923 cells, every wind term rested on that partial field, and the models were fitted on the cells it covered. It now covers all of them (section 2.2).
3. A terrain wind model, WindNinja, was run from the reanalysis over the site at 90 m for every flight hour of ten summers, over the surrounding 60 by 80 km box at 500 m, over the whole Purcell Trench at 1 km with hourly values kept at four valley stations, and over the site at 30 m and the lidar terrain at 10 m once per flight window. It was checked against the stations hour by hour. At Creston the station recorded 9.1 km/h over the flight hours, the reanalysis cell over it 5.8 and the model 3.2, with direction errors of 61 and 59 degrees, so the direction is the reanalysis's and the speed carries a constant step inside the model. The step was measured on flat ground at every mesh and removed, the direction limit is stated in the Methods (section 2.3, Table S10), and the fields entered the models beside the station wind, with the wind of the same period of the previous summer as well as the same period's, because the red crowns of a period record the flight of the summer before.

The spatial statistics were added beside the submitted models. Join counts and a Moran's I correlogram described the pattern of attack on each annual map before any model was fitted. The models were then built in the order of Aukema et al. (2008), with the terms for attack in and around each cell the year before chosen first, and the final annual model was refitted with a latent spatial

field and at grains of 90, 270 and 990 m (section 2.4, equations 1 to 3). Every candidate variable was screened beside the dependence terms by likelihood ratio test, variance inflation and an elastic net (Table 1), and the sixteen-day models and every further specification are reported in full in the Supporting Information (Tables S4, S11 and S12).

Findings

The Results (section 3) now answer the three questions in turn, each section naming the variables that bear on the question with their estimates, and the Discussion follows the same order. Three findings held. Stand density acted through the amount of host, with basal area entering at +0.430 log-odds per standard deviation in the submitted manuscript and at +0.276 in the revised final model, +0.167 with the terms for previous attack and +0.200 under the spatial field, so the thinnest stands had the least attack. Attack still peaked in stands of 25 to 30 cm quadratic mean diameter and fell in stands of larger trees, which held little pine. Ground facing the afternoon wind still had the least attack, with the wind shelter index entering at -0.267 before and at -0.073 now, -0.032 with the terms for previous attack ($p = 0.043$), and with no interaction with basal area ($p = 0.971$), as deposition predicts; the shelter index at each cell's own modelled wind direction, the share of flight hours blowing upslope and the slope updraft all pointed the same way (section 3.3).

Three findings changed. First, the submitted manuscript read shading as unsupported because north-facing ground had more attack, +0.384 before and +0.073 now. The revised manuscript reads the shade mechanism through the terms that measure shade by terrain, sky view, +0.285 before and +0.170 now with the terms for previous attack, and the direct sun of the flight window, +0.085, so terrain-shaded ground had less attack as the hypothesis predicted while north faces remained the exception, and it states that the effect did not survive the spatial field, under which sky view fell to +0.037 ($p = 0.285$). Second, the submitted manuscript tested plume disruption with the station wind and standing volume, an interaction of -0.017, which could no longer be distinguished from zero once the previous period's attack entered ($p = 0.296$). With the terrain wind of the previous summer and crown closure as the measure of openness, the interaction was +0.071 ($p < 0.001$), so that the wind effect was -0.085 in stands one standard deviation below the mean crown closure and +0.056 one above it. Wind lowered attack where the canopy was open and raised it where it was closed, which supports plume disruption in the sixteen-day models, while in the annual models the station wind raised attack at every density, +0.224 (section 3.2). Third, the shelter effect, although it held in sign, was small and changed sign under the latent spatial field, to +0.096 ($p < 0.001$), and lost significance at a 90 m grain, so the revised manuscript

reports it as weak and dependent on grain and place. The submitted abstract concluded, “Refugia are therefore proposed to lie on warm, windward ground in thin stands during windy flight periods.” The revised abstract concludes that refugia on this landscape were stands with little host, and that terrain and wind added small, scale-dependent modifiers to an outbreak whose spread was mostly contagion.

The added spatial statistics produced the result the revised manuscript leads with. Attack was clustered in every year, with 3.2 to 5.8 times as many pairs of neighbouring attacked cells as random permutations produced, and with positive autocorrelation to a median of 2,640 m. Attack in each cell and within 150 m the year before dominated every model it entered, raising the area under the receiver operating characteristic curve of the final annual model from 0.814 to 0.890. Residual Moran’s I remained significant in every year, at a median of 0.345 even under the spatial field, so the p-values of every model were optimistic, and section 4.3 says so.

Every number in the manuscript is computed by code in the manuscript file itself, and the file, the derived data and the analysis code are public at *https://github.com/seamusrobertmurphy/beetle-topography-and-wind-study*.

Yours sincerely,

Dr Seamus Murphy